

Daily Problem: 01–Feb–2026

Problem Statement

Over the alphabet $\{a, b\}$, let

$$E = \{ w \in \{a, b\}^* \mid \#_a(w) = \#_b(w) \}.$$

1. Show that E is not regular.
2. Using the answers of problems from Jan 19, Jan 28, and Feb 1, derive a contradiction to the claim that there are only finitely many prime numbers.

Part 1: The language E is not regular

Assume for contradiction that E is regular. Then the pumping lemma (Jan 29) applies to E . So there exists a pumping length $p \geq 1$ such that every word $w \in E$ with $|w| \geq p$ can be written as $w = xyz$ satisfying:

$$|xy| \leq p, \quad |y| \geq 1, \quad \text{and} \quad xy^kz \in E \text{ for all } k \geq 0.$$

Consider the word

$$w = a^p b^p.$$

Clearly $w \in E$ and $|w| = 2p \geq p$. By the pumping lemma, write $w = xyz$ with $|xy| \leq p$ and $|y| \geq 1$. Since the first p symbols of w are all a 's, the condition $|xy| \leq p$ forces both x and y to consist only of a 's. In particular, $y = a^t$ for some $t \geq 1$.

Now pump down with $k = 0$. The lemma claims $xz \in E$. But

$$xz = a^{p-t} b^p,$$

so

$$\#_a(xz) = p - t \neq p = \#_b(xz),$$

which implies $xz \notin E$. This contradicts the pumping lemma.

Therefore E is not regular.

Part 2: Contradiction if there are finitely many primes

Step 1 (Jan 19): Each L_n is regular

For each $n \in \mathbb{N}$, define

$$L_n = \{ w \in \{a, b\}^* \mid \#_a(w) - \#_b(w) \equiv 0 \pmod{n} \}.$$

From Jan 19, L_n is regular: there is a DFA with states $\{0, 1, \dots, n-1\}$ that tracks the value of $\#_a(w) - \#_b(w)$ modulo n .

Step 2 (Jan 28): The intersection over all primes equals E

Jan 28 establishes the identity

$$\bigcap_{p \text{ prime}} L_p = E.$$

(Intuitively: if $\#_a(w) - \#_b(w)$ is divisible by every prime, then it must be 0.)

Step 3: Assume finitely many primes and derive a contradiction

Assume, for contradiction, that there are only finitely many primes:

$$p_1, p_2, \dots, p_m.$$

Consider the integer

$$N = p_1 p_2 \cdots p_m.$$

Then for any integer d ,

$$(p_i \mid d \text{ for all } i = 1, \dots, m) \iff N \mid d,$$

because the primes p_i are pairwise coprime. Translating this into languages, we get

$$\bigcap_{i=1}^m L_{p_i} = L_N.$$

By Step 1, L_N is regular. Hence $\bigcap_{i=1}^m L_{p_i}$ is regular.

But under our assumption that these $p_1, \dots, p_m$ are *all* primes, we have

$$\bigcap_{p \text{ prime}} L_p = \bigcap_{i=1}^m L_{p_i},$$

so the left-hand side would be regular as well.

Now apply Step 2 (Jan 28):

$$\bigcap_{p \text{ prime}} L_p = E.$$

Therefore E would be regular.

This contradicts Part 1, where we proved E is not regular.

Conclusion

Our assumption that there are finitely many primes leads to a contradiction. Hence there must be infinitely many prime numbers.